

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	31 January 2025
Team ID	SWTID-2026-9682
Project Name	AI-Augmented Backend Development – AI StudyBuddy API
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The four team members come together, agree the goal of the session and choose one problem statement to focus on. It is framed as a “How might we” question, based on PS-1 to PS-3.

Team gathering

Team ID: SWTID-2026-9682

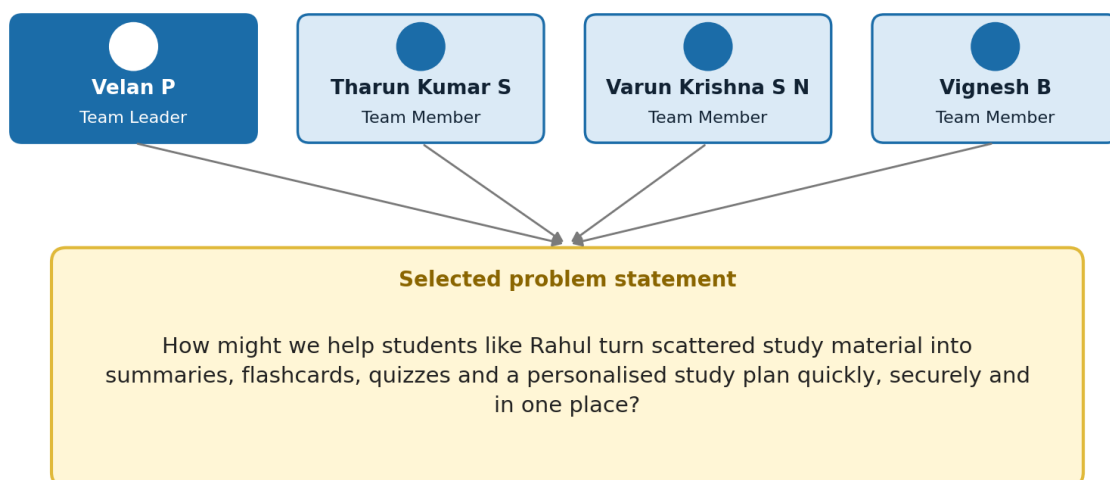


Figure 1: Team and selected problem statement

Step-2: Brainstorm, Idea Listing and Grouping

Ideas are listed freely first, then grouped into four themes. Dashed notes are future ideas that sit outside the current project scope.

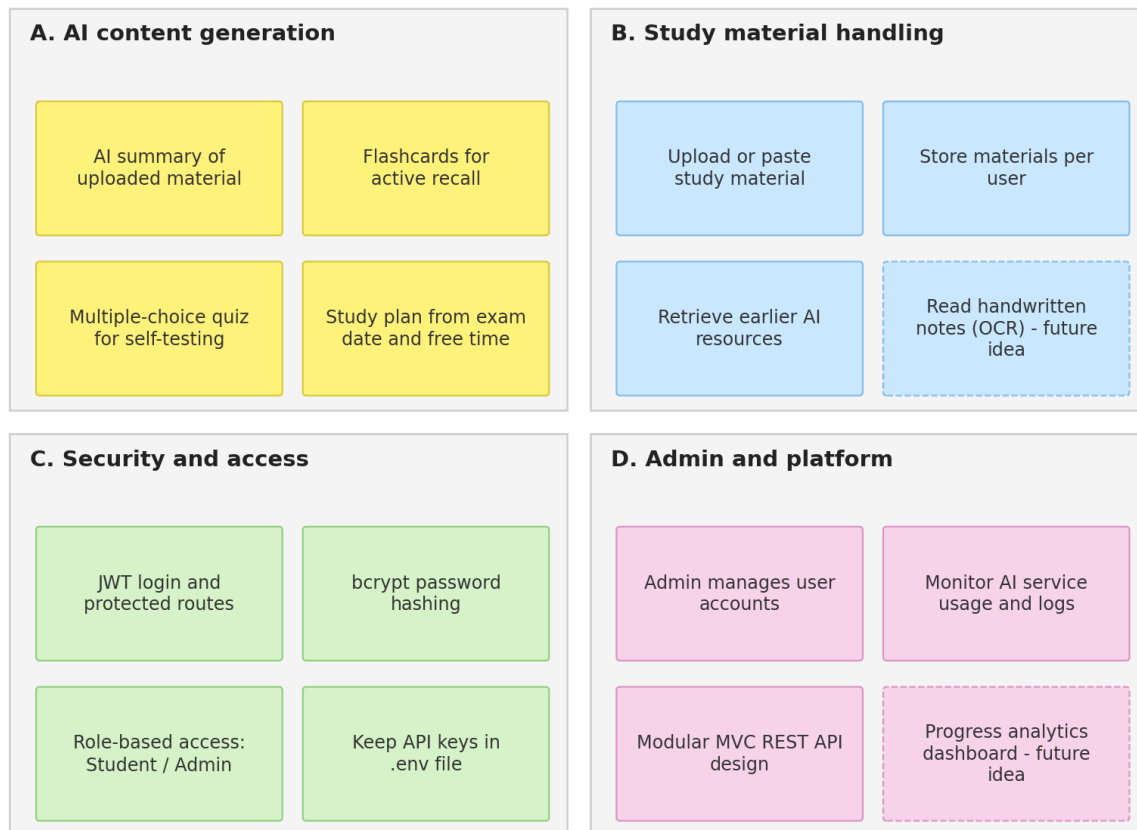


Figure 2: Ideas grouped by theme

Step-3: Idea Prioritization

Each idea is placed on an importance versus feasibility grid. Importance is how much the idea helps a student like Rahul. Feasibility is how easily the team can build it with Node.js, Express.js, MongoDB and the Google Gemini API.

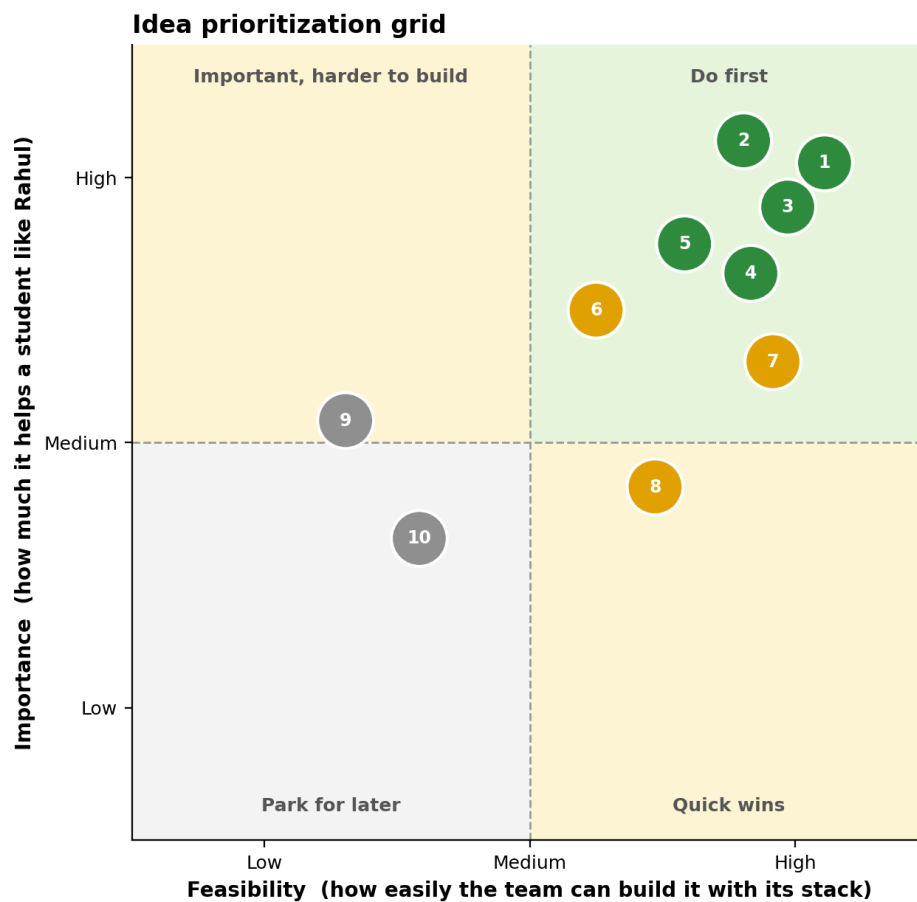


Figure 3: Prioritization grid (green = build first, amber = build next, grey = future scope)

#	Idea	Importance	Feasibility	Priority
1	JWT login and bcrypt password hashing	High	High	Build first
2	AI summary generation	High	High	Build first
3	Study material upload and storage	High	High	Build first
4	Flashcard generation	High	High	Build first
5	Quiz generation	High	Medium	Build first
6	Personalised study plan	Medium	Medium	Build next
7	Role-based access (Student / Admin)	Medium	High	Build next
8	Admin monitoring and logs	Low	Medium	Build next
9	OCR for handwritten notes (future idea)	Medium	Low	Future scope
10	Progress analytics (future idea)	Low	Low	Future scope

References:

- Mural – Brainstorm and idea prioritization template: <https://mural.co/templates/ideate>
- Google – Gemini API text generation: <https://ai.google.dev/gemini-api/docs/text-generation>
- JWT.io – Introduction to JSON Web Tokens: <https://jwt.io/introduction>
- Mongoose – Schemas guide: <https://mongoosejs.com/docs/guide.html>